Performance Evaluation of ML and DL Models for EEG-Based Stress Detection Using the DASPS Dataset

Goldy Saini
Department of Computer Science & Engineering
Rabindranath Tagore University,
Bhopal, MP, India
goldysainibe@gmail.com

Rakesh Kumar
Department of Computer Science & Engineering
Rabindranath Tagore University,
Bhopal, MP, India
rakeshmittan@gmail.com

Lokesh Malviya
School of Computing Science & Engineering
VIT Bhopal University
Kothrilan, MP, India
lokeshmalviya@vitbhopal.ac.in

Divya Thakur
School of Computer Science and Engineering
Manipal University Jaipur,
Jaipur, Rajasthan, India
divya.thakur@jaipur.manipal.edu

2025 International Conference on Engineering Innovations and Technologies (ICoEIT) | 979-8-3315-2595-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICOEIT63558.2025.11211768

***Abstract*— EEG-based stress detection has become a vital area of research for mental health monitoring and real-time cognitive assessment. Aiming to classify stress using electroencephalogram (EEG) data, this study will compare and contrast ML and DL models. By using dimensionality reduction techniques (PCA and Fast-ICA) to several EEG frequency bands (Theta, Alpha, and Beta), we want to evaluate the performance of DL architectures (LSTM, GRU, and Hybrid LSTM-GRU) compared to typical ML models (SVM, Random Forest, AdaBoost, Gradient Boosting).The DASPS dataset's pre-processed EEG data was feature extracted using PCA and Fast-ICA after Butterworth filtering to remove noise. Whereas DL models handled time-series EEG data, capturing sequential dependencies, ML models were trained on structured features. Accuracy, precision, recall, and loss measures guided evaluation of performance. With GRU achieving the highest accuracy (99.45%) and the lowest loss (0.0419) with Fast-ICA, deep learning models beat conventional ML models, according results. Random Forest obtained the best accuracy (98.6%) among ML models; yet, GRU exceeded it. This work finds that deep learning—especially GRU—offers better performance in EEG-based stress detection, hence perfect for real-time applications.**

Keywords— EEG-based stress detection, Machine Learning, Deep Learning, GRU, Feature Extraction (PCA & Fast-ICA)

I. Introduction

A common problem influencing people all around, stress causes several psychological and physical problems including anxiety, sadness, heart ailments, and poor cognitive ability. Developing sensible treatments and customised healthcare solutions depends on precise stress detection. An emerging technique for stress detection is electroencephalography (EEG), which is non-invasive, has high temporal resolution, and can record brain reactions associated with emotional and cognitive states brought on by stress[1]–[5]. Conventional methods of stress detection rely on physiological markers such heart rate variability (HRV), galvanic skin response (GSR), personally as well reported evaluations. A reliable tool for stress measurement, however, is electroencephalography (EEG), since it is a more direct indication of brain activity. The efficiency and accuracy of stress detection based on electroencephalograms (EEGs) have been substantially improved by recent advances in deep learning. Deep learning models—including convolutional neural networks (CNNs), long short-term memory (LSTM) networks, and hybrid models—have outperformed more traditional machine learning approaches. By automatically learning hierarchical features from EEG data, these methods improve classification accuracy while decreasing the requirement for human feature engineering. Turning raw EEG signals into meaningful representations depends on feature extraction, which also helps the models to identify stress-related patterns from normal brain activity[6]–[10].

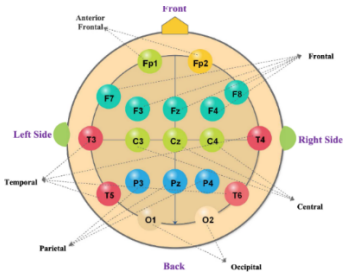

Fig. 1 Stress detection using EEG[11]

Approaches in the time-domain, frequency-domain, and time-frequency domain have all been investigated as potential feature extraction methods for EEG-based stress detection.

979-8-3315-2595-8/25/$31.00 ©2025 IEEE
DOI: 10.1109/ICoEIT.2025.48

Authorized licensed use limited to: VIT University. Downloaded on March 26,2026 at 13:54:47 UTC from IEEE Xplore. Restrictions apply.

Mean, variance, skewness, and kurtosis are time domain features that capture EEG signals' statistical properties; features obtained through spectral analysis techniques like Wavelet Transform or Fast Fourier Transform (FFT) are frequency domain features that reveal the power distribution of various brainwave bands (gamma, beta, alpha, theta, or delta). Continuous Wavelet Transform (CWT) or Short-Time Fourier Transform (STFT) are two time-frequency methods that preserve spectral and temporal information, which allows them to describe EEG data more comprehensively [12]– [14].

Examining the relative merits of deep learning models or feature extraction methods for EEG-based stress detection is the driving force behind this study. Using EEG data, we specifically assess CNNs, LSTMs, and hybrid models in their ability to classify stress degrees. Moreover, under study are the consequences of numerous feature extraction techniques on model correctness, resilience, and computing economy. The output of this effort will provide perceptive study of the ideal combination of deep learning architectures and feature extraction methods for uses in real-time stress monitoring. By bridging the gap between neuroscience, artificial intelligence, and biomedical engineering, this work aids to design intelligent stress detection systems that might be integrated into wearable gadgets, mobile apps, and healthcare monitoring systems. The purpose of this comparison study is to promote the use of deep learning technologies in wellness and mental health applications and to improve the field of EEG-based stress detection.

II. LITERATURE REVIEW

Mhaouch et al. (2024) [15] designed an automated system for stress detection using an electroencephalogram (EEG) and a Recurrent Neural Network (RNN) model, with a focus on the Gated Recurrent Unit (GRU). Preprocessing Fast Fourier Transform (FFT) on EEG signals from Kaggle yielded an accuracy of 99.53% in the study, thanks to the Adam optimiser. Confusion matrices and accuracy-loss curves suggest that the outcomes showed consistent stress emotion detecting with a minimum error rate.

Luo et al. (2024) [16] proposed an algorithm for quantifying and assessing GAD using electroencephalogram (EEG) data and ensemble learning. Utilising functional connectivity features, data from 80 GAD patients and 39 normal adults were analysed, with each group divided into four severity groups. The CatBoost model achieved 98.1% accuracy when classifying GAD levels using a 10-second time window. The research showed that the temporal and frontal lobes have significant functional connections, particularly during the beta rhythm.

Hosseini et al. (2023) [1] deep learning-based emotion identification system combining the Big Five personality features with EEG inputs. Using a pre-trained CNN for feature extraction, EEG data from 60 people were examined; an LSTM network predicted personality traits. Then, a novel classifier accurately predicted arousal and valence emotional states with a rate of 93.97 percent.

Tripathy et al. (2023) [10] proposed the FBDAWT as a method for anxiety detection in wearable electrocardiogram sensors. Utilising XGBoost, we were able to classify ECG data into four levels of anxiety: normal, mild, moderate, and severe. The model attained an accuracy of 92.27%, an F1-score of 92.13%, and a Kappa score of 0.89 by means of a publically available dataset. Emphasising its fit for real-time wearable health monitoring applications, the study contrasted Facebook DAWT with current techniques to show its performance in physiological signal-based anxiety identification.

III. RESEARCH METHODOLOGY

This study recorded from 14 electrodes, uses EEG data from the DASPS dataset and groups into Theta, Alpha, and Beta frequency regions. Preprocessing calls for segmentation, normalising, and Butterworth filter noise reduction. Dimensionality is lowered via feature extraction applied with PCA and Fast-ICA. Whereas deep learning models (LSTM, GRU, hybrid LSTM-GRU) analyse time-series EEG data, machine learning models (SVM, Random Forest, AdaBoost, Gradient Boost) are trained on structured features. Using the Adam optimiser, models are optimised and assessed using accuracy, precision, recall, and loss, therefore guaranteeing a thorough comparison of stress categorisation methods.

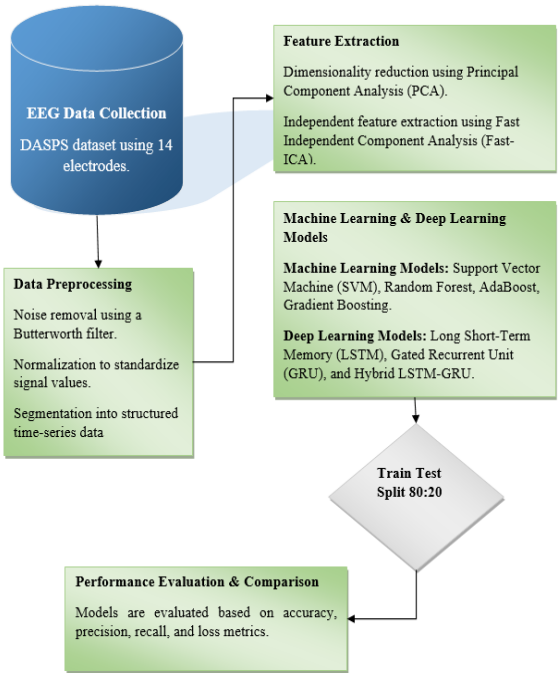

Fig. 2 Proposed Flowchart

A. *Data Collection*

This work makes use of DASPS dataset EEG data, which comprises brainwave recordings from 14 electrodes positioned following the 10-20 international system. Based on valence and arousal scores, the dataset comprises low,

Authorized licensed use limited to: VIT University. Downloaded on March 26,2026 at 13:54:47 UTC from IEEE Xplore. Restrictions apply.

moderate, and high anxiety levels of EEG signals obtained from individuals various degrees of stress. By sampling the EEG signals at 128 Hz, brain activity is high-resolution captured. The theta (4-8 Hz), alpha (8-12 Hz), and beta (12-30 Hz) bands are the main ones used for subject data segmentation. These ranges are known to coincide with different mental states, such as stress, relaxation, or concentration. Every EEG segment is labelled based on stress levels to guarantee enough data for model training, therefore rendering the dataset fit for methods of supervised learning. Strong capture of sequential correlations in brainwave activity is made possible by preprocessing and structural analysis of the dataset, which produce time-series EEG data [24].

B. Data Pre-processing

Retaining basic brain activity patterns, reducing noise, and guaranteeing data quality all depend on pre-processing EEG signals. The first stage is artefact reduction using a Butterworth bandpass filter to eliminate unwanted noise from muscle actions, eye blinks, and electrical interference thereby maintaining just relevant frequency ranges. Then min-max scaling enables each EEG channel to be normalised such that all signals fall within a comparable range, therefore preventing bias during model training. Deep learning algorithms demand ordered sequential data, hence splitting EEG signals into overlapping time periods helps to keep temporal relationships and guarantees a balanced dataset [25]. The dataset is split then in an 80–20 ratio: 80% for testing and 20% for training. In deep learning models, sequence padding also helps to standardise input lengths so they may efficiently process EEG sequences. This pre-processing technique guarantees excellent signal clarity, consistent input formats, and improved model performance.

C. Feature Extraction

One way to decrease dimensionality while keeping the most significant stress-related EEG data is by using features extraction. The two main methodologies that are employed are Principal Component Analysis (PCA) and Fast Independent Component Analysis (Fast-ICA). By minimising data duplication while preserving key EEG patterns, principal component analysis (PCA) finds orthogonal components that maximise variance. On the other hand, Fast-ICA separates biologically important patterns connected with stress responses by isolating independently arising independent components in EEG data. Two modelling techniques then benefit from the obtained features:

- For Machine Learning Models: Classifiers like SVM, Random Forest, AdaBoost, or Gradient Boosting take extracted EEG features as input.
- For Deep Learning Models: The original EEG time-series data is used to automatically train a Hybrid LSTM-GRU model, which consists of a Gated Recurrent Unit (GRU) & Long Short-Term Memory (LSTM), to recognise patterns linked to stress [25].

These techniques ensure that only the informative EEG features are used, improving classification accuracy while minimizing computational overhead [26].

D. EDA

The distribution of EEG signals across several stress levels is investigated using EDA. To investigate EEG signal patterns and identify outliers, first histograms and boxplots are created. Variations in frequency band are visualized using spectrograms and power spectral density (PSD) plots, therefore offering understanding of how stress levels affect brain activity. By means of correlation heatmaps, correlations between stress intensity and EEG channels are revealed, therefore guiding the most important electrodes in stress classification. Time-series plots are produced to investigate EEG variations under various stress levels, therefore supporting the validation of the choice of deep learning models able to efficiently detect temporal relationships. confirm by EDA that whilst Theta and Alpha waves are more prevalent in relaxed situations, Beta band activity is raised under high-stress conditions [27]. These results offer important new perspectives on EEG signal behavior and direct model choice towards best stress detection performance.

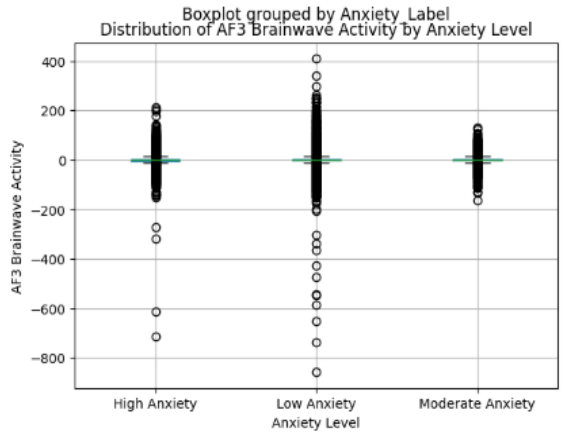

Fig. 3 Boxplot Grouped by Stress label Distribution

Three Stress levels—High, Low, and Moderate—shown on a boxplot graph (Fig. 3) divide the AF3 brainwave activity. Every category reveals median values and variability; however, it is important to underline how distinct brainwave activity depends on stress level. This paper highlights the relationship between brainwave cycles and stress.

Authorized licensed use limited to: VIT University. Downloaded on March 26,2026 at 13:54:47 UTC from IEEE Xplore. Restrictions apply.

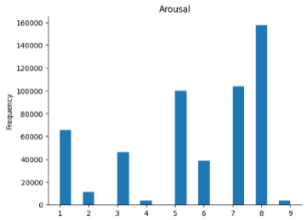

Fig. 4 Histogram plot of Frequency and Arousal

The histogram (Fig. 4) presents the several frequency ranges of arousal degree. At arousal level 8, the frequency is maximal; subsequently, it follows by 5 and 7. Levels 1 and 2 have the lowest frequency; levels 3, 4, 6, as well as 9 have rather moderate frequency.

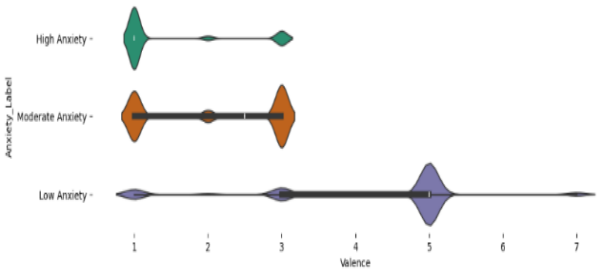

Fig. 5 Boxplot of Stress label

Stress Label Box Plot (Fig. 5) depicts the distribution of low, moderate, high stress levels over different valence scores 1 to 7. The horizontal axis displays valence; the vertical axis shows stress levels.

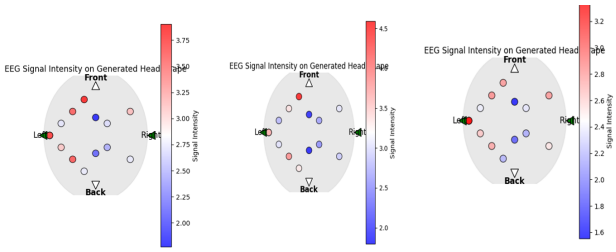

Fig. 6 EEG Signal Intensity Map Showing Brain Activity Distribution

Using colour-coded circles to indicate varying strengths of the EEG signals (Fig. 6), the top-down view of a head displayed by the image intensity shows Labels for the front, back, left, and right aid the viewer; a colour bar on the right indicates the intensity scale, ranging from 2.25 to 3.75, therefore demonstrating differences in brain activity across multiple areas. Analogously, the EEG signal intensity map graphically illustrates the electrical activity in the brain by exhibiting signal strength variations over several sites. Important for understanding cognitive processes, the colour-coded gradients highlight these changes and help to evaluate areas of either higher or lower brain activity. Emphasising even more the distribution of electrical signals, another EEG intensity map graphically distinguishes areas of strong and weak activity by colour variations.

E. Machine Learning & Deep Learning Modeling

Evaluate EEG-based stress detection by comparing the performance of deep learning models with machine learning models in identifying stress levels.

Machine Learning Models:

- **Support Vector Machine (SVM):** A reliable classifier that determines the best tolerance levels for stress.
- **Random Forest:** An ensemble-based model that improves classification through multiple decision trees.
- **AdaBoost & Gradient Boosting:** Boosting algorithms that enhance weak classifiers for improved prediction accuracy.

Deep Learning Models

- **Long Short-Term Memory (LSTM):** May experience problems with vanishing gradients, but effectively captures long-term interdependence in EEG sequences.
- **Gated Recurrent Unit (GRU):** A computationally efficient alternative to LSTM, mitigating vanishing gradients while retaining sequential memory [25].
- **Hybrid LSTM-GRU Model:** Takes advantage of GRU's computational efficiency and LSTM's long-term memory retention capabilities [25][28].

IV. RESULT & DISCUSSION

Combining recall, loss, accuracy, and precision, many ML and DL models were tested for stress detection based on EEG data. Whenever it comes to feature extraction methods or frequency ranges, deep learning models consistently outperform more traditional machine learning models.

$$Accuracy = \frac{(TP+TN)}{(TP+FP+TN+FN)} \quad (1)$$

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

$$Loss = -\frac{1}{m}\sum_{i=1}^{m} \mathcal{Y}i.\log(\mathcal{Y}i) \quad (4)$$

TABLE I PERFORMANCE COMPARISON TABLE

Model	Feature Extraction	Accuracy (%)	Precision (%)	Recall (%)	Loss
SVM	PCA	91.1	92.5	89.8	0.0852
Random Forest	PCA	98.6	98.9	97.2	0.0456
AdaBoost	PCA	97.9	98.2	96.5	0.0521

Authorized licensed use limited to: VIT University. Downloaded on March 26,2026 at 13:54:47 UTC from IEEE Xplore. Restrictions apply.

Gradient Boosting	PCA	98.3	98.6	96.8	0.0489
LSTM	Fast-ICA	98.21	98.81	95.85	0.0604
GRU	Fast-ICA	**99.45**	**99.52**	**97.15**	**0.0419**
Hybrid LSTM-GRU	Fast-ICA	98.07	99.01	95.51	0.0583

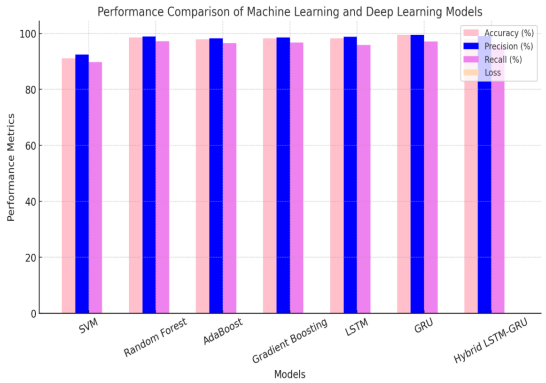

Fig. 7 Performance Comparison graph of Machine learning and Deep Learning Models Gra

A. Performance of Machine Learning Models

Classical ML models, such as SVM, Random Forest, AdaBoost, & Gradient Boosting, demonstrated excellent classification capabilities. Figure 7 and Table I show that the most effective methods, Random Forest & Gradient Boosting, achieved an accuracy of up to 98.6%.Since these models relied mostly on synthetic characteristics obtained from PCA and Fast-ICA, they were less adaptable to changes in real-time EEG signals.

B. Performance of Deep Learning Models

Sequential dependencies in EEG signals were used by deep learning models—especially GRU, LSTM, and Hybrid LSTM-GRU—to raise stress detection accuracy. Achieving 99.45% accuracy with Fast-ICA feature extraction and 99.32% with PCA, GRU emerged as the best-performing model (Fig. 7 and Table 2). Its lowest loss values also make it the most reliable and efficient model for EEG-based stress categorisation. Although the hybrid LSTM-GRU model showed better recall and precision than GRU, its loss values were higher. Although LSTM performed steadily, it was less effective in managing vanishing gradients, which resulted in much reduced accuracy compared to GRU.

C. Comparison of PCA and Fast-ICA

Models' performance was substantially influenced by feature extraction. Fast-ICA regularly beat PCA, resulting in reduced loss values all around and improved classification accuracy. Fast-ICA was very successful in improving GRU function, therefore enabling more effective collection of stress-related EEG signals. Although PCA helped to lower dimensionality, occasionally it resulted in information loss, therefore somewhat compromising classification accuracy.

D. Overall Discussion

Based on the results, GRU is the top model for EEG-based stress detection since it reduces disappearing gradients and effectively analyses sequential EEG data. Although traditional machine learning models performed well, their limited practicality for real-time applications stems from their need for intensive feature engineering. Although the hybrid LSTM-GRU model produced competitive results with somewhat higher loss values, GRU is the recommended model for pragmatic applications. The results imply that to increase EEG-based stress detection even more, next studies should investigate GRU-based architectures with extra optimisations including attention mechanisms.

With the best accuracy and lowest loss values, the study verifies that GRU is the most successful model for EEG-based stress detection. Fast-ICA feature extraction is the advised approach for dimensionality reduction. Deep learning approaches provided more flexibility, but standard ML models—Random Forest and Gradient Boosting in particular—achieved high accuracy. Although the hybrid LSTM-GRU model showed somewhat greater loss values, it performed well in precision and recall, hence GRU is the most appropriate model for real-time EEG stress detection uses. By means of attention mechanisms and reinforcement learning, future work can develop GRU-based designs to raise classification accuracy and efficiency even further.

TABLE II COMPARATIVE ANALYSIS BETWEEN EXISTING AND PROPOSED MODEL

Models	Accuracy	Precision	Recall	Ref.
DCNN-LSTM	95.25	98.00	97.00	[19]
SVM	85.20	--	--	[22]
RBF-SVM and kNN yielded	79.81	--	--	[23]
Proposed GRU	**99.45**	**99.52**	**97.15**	--

Authorized licensed use limited to: VIT University. Downloaded on March 26,2026 at 13:54:47 UTC from IEEE Xplore. Restrictions apply.

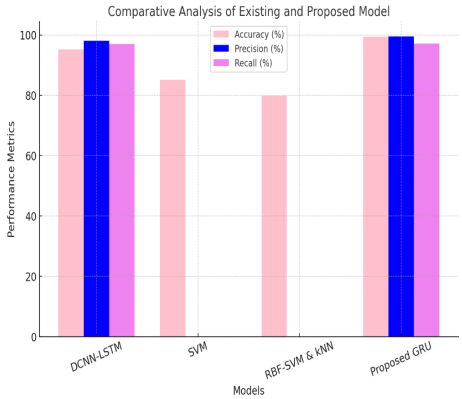

Fig. 8 Comparative Analysis Graph

Table II and Fig. 8 compares the proposed GRU-based model for EEG-based stress detection with current models. The Deep Convolutional Neural Network-Long Short-Term Memory (DCNN-LSTM) model did well, although it was only 95.25% accurate, whereas the suggested model was 98.00% accurate. Support Vector Machine (SVM) models rely on handmade features instead of deep learning-based hierarchical feature extraction, which results in lower classification capabilities and possible misclassifications (with an accuracy of 85.20%). Consequently, these models lack precision and recall values. Analogous to this, the RBF-SVM and kNN models obtained an accuracy of 79.81%, much below the suggested model, implying restrictions in managing complexity of EEG signals. The proposed Gated Recurrent Unit model outperformed all previous models with best accuracy of 99.45%, 99.52%, and 97.15%. Better performance of GRU can be explained by its ability to effectively capture sequential dependencies in EEG signals and minimise vanishing gradient difficulties. Moreover, Fast-ICA for feature extraction helped to keep essential stress-related patterns, so enhancing classification accuracy. These findings indicate that among real-time EEG-based stress detection models, GRU is the most consistent and powerful one.

V. CONCLUSION

In conclusion, by contrasting conventional ML models with more sophisticated deep learning (DL) architectures, this study sought to improve EEG-based stress detection. The goals were to compare the classification performance of DL models (LSTM, GRU, Hybrid LSTM-GRU) with that of ML models (SVM, Random Forest, AdaBoost, Gradient Boost), and to determine the effect of feature extraction methods (PCA and Fast-ICA) on classification accuracy. The technique comprised segmenting the pre-processed EEG data from the DASPS dataset into Theta, Alpha, and Beta frequency bands after Butterworth filter noise reduction. Using PCA and Fast-ICA, features were retrieved; sequential data was handled by DL models while structured data fed into ML models. Trained on the Adam optimiser, models were assessed using loss, accuracy, precision, and recall measures. With Fast-ICA, the results showed that GRU exceeded all models with the best accuracy (99.45%) and the lowest loss (0.0419). Though it demanded significant feature engineering, Random Forest was the best-performing ML model (98.6% accuracy). The work validates that deep learning—especially GRU—is the most efficient model for real-time EEG-based stress detection, hence fit for wearable devices and hospital monitoring. Future studies should look into GRU-based designs including attention mechanisms for better performance.

REFERENCES

[1] M. S. K. Hosseini, S. M. Firoozabadi, K. Badie, and P. Azadfallah, "Personality-Based Emotion Recognition Using EEG Signals with a CNN-LSTM Network," Brain Sci., vol. 13, no. 6, 2023, doi: 10.3390/brainsci13060947.

[2] R. Mohan and S. Perumal, "Classification and Detection of Cognitive Disorders like Depression and Anxiety Utilizing Deep Convolutional Neural Network (CNN) Centered on EEG Signal," Trait. du Signal, vol. 40, no. 3, pp. 971–979, 2023, doi: 10.18280/ts.400313.

[3] Y. Tao, M. Yang, H. Shen, Z. Yang, Z. Weng, and B. Hu, "Classifying Anxiety and Depression through LLMs Virtual Interactions: A Case Study with ChatGPT," Proc. - 2023 2023 IEEE Int. Conf. Bioinforma. Biomed. BIBM 2023, no. June, pp. 2259–2264, 2023, doi: 10.1109/BIBM58861.2023.10385305.

[4] N. B. Ogur, C. Ceken, Y. S. Ogur, H. U. Yuvaci, A. B. Yazici, and E. Yazici, "Development of an Artificial Intelligence-Supported Hybrid Data Management Platform for Monitoring Depression and Anxiety Symptoms in the Perinatal Period: Pilot-Scale Study," IEEE Access, vol. 11, no. February, pp. 31456–31466, 2023, doi: 10.1109/ACCESS.2023.3262467.

[5] A. Ksibi, M. Zakariah, L. J. Menzli, O. Saidani, L. Almuqren, and R. A. M. Hanafieh, "Electroencephalography-Based Depression Detection Using Multiple Machine Learning Techniques," Diagnostics, vol. 13, no. 10, 2023, doi: 10.3390/diagnostics13101779.

[6] S. Q. O. OMAR and C. TEPE, "EEG İşareti tabanlı anksiyete sınıflandırması için Dalgacık dönüşümü ile öznitelik çıkarma," Ömer Halisdemir Üniversitesi Mühendislik Bilim. Derg., vol. 12, no. 3, pp. 726–732, 2023, doi: 10.28948/ngumuh.1230092.

[7] A. Al-Ezzi et al., "Machine learning for the detection of social anxiety disorder using effective connectivity and graph theory measures," Front. Psychiatry, vol. 14, 2023, doi: 10.3389/fpsyt.2023.1155812.

[8] C. Jang et al., "The impact of comorbid anxiety on quantitative EEG heterogeneity in children with attention-deficit/hyperactivity disorder," Front. Psychiatry, vol. 14, no. July, pp. 1–11, 2023, doi: 10.3389/fpsyt.2023.1190713.

[9] C. Vidaurre, V. V. Nikulin, and M. Herrojo Ruiz, "Identification of spatial patterns with maximum association between power of resting state neural oscillations and trait anxiety," Neural Comput. Appl., vol. 35, no. 8, pp. 5737–5749, 2023, doi: 10.1007/s00521-022-07847-5.

[10] R. K. Tripathy, D. K. Dash, S. K. Ghosh, and R. B. Pachori, "Detection of Different Stages of Anxiety From Single-Channel Wearable ECG Sensor Signal Using Fourier-Bessel Domain Adaptive Wavelet Transform," IEEE Sensors Lett., vol. 7, no. 5, 2023, doi: 10.1109/LSENS.2023.3274668.

[11] L. Malviya and S. Mal, "A novel technique for stress detection from EEG signal using hybrid deep learning model," Neural Comput. Appl., vol. 34, no. 22, pp. 19819–19830, Nov. 2022, doi: 10.1007/S00521-022-07540-7/METRICS.

[12] N. Jin, Y. Wu, J. Park, Z. Qin, and H. N. Liang, "Brain-Metaverse Interaction for Anxiety Regulation," 2023 9th Int. Conf. Virtual Reality, ICVR 2023, no. May, pp. 385–392, 2023, doi: 10.1109/ICVR57957.2023.10169785.

[13] U. M. Haque, E. Kabir, and R. Khanam, "Early detection of paediatric and adolescent obsessive–compulsive, separation anxiety and attention

Authorized licensed use limited to: VIT University. Downloaded on March 26,2026 at 13:54:47 UTC from IEEE Xplore. Restrictions apply.

deficit hyperactivity disorder using machine learning algorithms," Heal. Inf. Sci. Syst., vol. 11, no. 1, pp. 1–14, 2023, doi: 10.1007/s13755-023-00232-z.

[14] Z. Shen et al., "Aberrated Multidimensional EEG Characteristics in Patients with Generalized Anxiety Disorder: A Machine-Learning Based Analysis Framework," Sensors, vol. 22, no. 14, 2022, doi: 10.3390/s22145420.

[15] A. Mhaouch, M. Fradi, W. Gtifa, A. Ben Abdelali, and M. Machhout, "Deep Learning Based Recurrent Neural Network Model for Stress Detection in EEG Signals," Rev. d'Intelligence Artif., vol. 38, no. 3, pp. 979–985, 2024, doi: 10.18280/ria.380323.

[16] X. Luo, B. Zhou, J. Fang, Y. Cherif-Riahi, G. Li, and X. Shen, "Integrating EEG and Ensemble Learning for Accurate Grading and Quantification of Generalized Anxiety Disorder: A Novel Diagnostic Approach," Diagnostics, vol. 14, no. 11, 2024, doi: 10.3390/diagnostics14111122.

[17] K. R. Pathak and F. Haneef, "EEG based Stress Analysis Through Feature Extraction," Int. J. Recent Innov. Trends Comput. Commun., vol. 11, no. 8, pp. 140–144, 2023, doi: 10.17762/ijritcc.v11i8.7931.

[18] A. A. Bamanikar, R. V. Patil, and L. V. Patil, "Stress Analysis using Feature Extraction Approach Using EEG Signal," Int. J. Intell. Syst. Appl. Eng., vol. 12, no. 5s, pp. 409–417, 2024.

[19] S. A. Patil and A. N. Paithane, "Optimized EEG-Based Stress Detection: A Novel Approach," Biomed. Pharmacol. J., vol. 17, no. 4, pp. 2607–2616, 2024, doi: 10.13005/bpj/3052.

[20] B. Roy et al., "Hybrid Deep Learning Approach for Stress Detection Using Decomposed EEG Signals," Diagnostics, vol. 13, no. 11, pp. 1–19, 2023, doi: 10.3390/diagnostics13111936.

[21] H. M. Afify, K. K. Mohammed, and A. E. Hassanien, "Stress detection based EEG under varying cognitive tasks using convolution neural network," Neural Comput. Appl., vol. 37, no. 7, pp. 5381–5395, 2025, doi: 10.1007/s00521-024-10737-7.

[22] S. M. U. Saeed, S. M. Anwar, H. Khalid, M. Majid, and U. Bagci, "EEG based classification of long-term stress using psychological labeling," Sensors (Switzerland), vol. 20, no. 7, pp. 1–15, 2020, doi: 10.3390/s20071886.

[23] A. Hemakom, D. Atiwiwat, and P. Israsena, "ECG and EEG based detection and multilevel classification of stress using machine learning for specified genders: A preliminary study," PLoS One, vol. 18, no. 9 September, pp. 1–24, 2023, doi: 10.1371/journal.pone.0291070.

[24] Ghonchi, H., Foulsham, T., & Ferdowsi, S. (2024, August). Assessing Neural Patterns of Anxiety Using Deep Learning: An EEG Study. In 2024 32nd European Signal Processing Conference (EUSIPCO) (pp. 1596-1600). IEEE.

[25] Roy B, Malviya L, Kumar R, Mal S, Kumar A, Bhowmik T, Hu JW. Hybrid Deep Learning Approach for Stress Detection Using Decomposed EEG Signals. Diagnostics. 2023; 13(11):1936. https://doi.org/10.3390/diagnostics13111936

[26] Chen, J. X., Jiang, D. M., & Zhang, Y. N. (2019). A hierarchical bidirectional GRU model with attention for EEG-based emotion classification. Ieee Access, 7, 118530-118540.

[27] Martínez Vásquez, D. A., Posada-Quintero, H. F., & Rivera Pinzón, D. M. (2023). Mutual information between EDA and EEG in multiple cognitive tasks and sleep deprivation conditions. Behavioral Sciences, 13(9), 707.

[28] Zafar, N., Haq, I. U., Chughtai, J. U. R., & Shafiq, O. (2022). Applying hybrid LSTM-GRU model based on heterogeneous data sources for traffic speed prediction in urban areas. Sensors, 22(9), 3348.

Authorized licensed use limited to: VIT University. Downloaded on March 26,2026 at 13:54:47 UTC from IEEE Xplore. Restrictions apply.